

61a.

Se quiere obtener

$$e^i \cdot e_j = \delta_{ij}$$

entonces para ortogonalidad

$$e^1 = \alpha(e_2 \times e_3)$$

de forma que

$$(e_2 \times e_3) \cdot e_2 = 0 \quad (e_2 \times e_3) \cdot e_3 = 0$$

ahora, si se toma

$$e^1 \cdot e_1 = \alpha(e_2 \times e_3) \cdot e_1 = \alpha(e_1(e_2 \times e_3))$$

de donde como se busca $e^1 \cdot e_1 = 1$ se ve que $\alpha = 1/V$ Por lo tanto

$$e^1 = \frac{e_2 \times e_3}{V}$$

$$e^2 = \frac{e_3 \times e_1}{V}$$

$$e^3 = \frac{e_1 \times e_2}{V}$$

de forma que

$$(e_2 \times e_3) \cdot e_2 = 0 \quad (e_2 \times e_3) \cdot e_3 = 0$$

ahora, si se toma

$$e^1 \cdot e_1 = \alpha (e_2 \times e_3) \cdot e_1 = \alpha (e_1 (e_2 \times e_3))$$

de donde como se busca $e^1 \cdot e_1 = 1$ se ve que $\alpha = 1/V$, por lo tanto

$$e^1 = \frac{e_2 \times e_3}{V}$$

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$$e^3 = \frac{e_1 \times e_2}{V}$$

se concluye que

$$e^i = \frac{e_j \times e_k}{e_i \cdot (e_j \times e_k)}$$

b) $V\tilde{V}=1$

$$[e_1 \cdot (e_2 \times e_3)][e^1 \cdot (e^2 \times e^3)] = 1$$

$$[(e_1)_i \varepsilon^{ijk} (e_2)_j (e_3)_k] [(e^1)_m \varepsilon^{mny} (e^2)_n (e^3)_y] = 1 \quad (e_1)_i (e^1)_m = \delta_{im}$$

$$\delta_{im} \varepsilon^{ijk} (e_2)_j (e_3)_k \varepsilon^{mny} (e^2)_n (e^3)_y = 1$$

$$\varepsilon^{ijk} (e_2)_j (e_3)_k \varepsilon^{iny} (e^2)_n (e^3)_y = 1$$

$$(\delta_n^j \delta_y^k - \delta_y^j \delta_n^k) (e_2)_j (e_3)_k (e^2)_n (e^3)_y = 1$$

$$e^1 = \frac{e_2 \times e_3}{V}$$

$$e^2 = \frac{e_3 \times e_1}{V}$$

$$e^3 = \frac{e_1 \times e_2}{V}$$

se concluye que

$$e^i = \frac{e_j \times e_k}{e_l \cdot (e_j \times e_k)}$$

$$b) \quad V \tilde{V} = 1$$

$$[e_1 \cdot (e_2 \times e_3)] [e^1 \cdot (e^2 \times e^3)] = 1$$

$$[e_i]_j \varepsilon^{ijk} (e_2)_j (e_3)_k [e^1]_m \varepsilon^{mny} (e^2)_n (e^3)_y = 1 \quad [e_i]_j [e^1]_m = \delta_{jm}$$

$$\delta_{im} \varepsilon^{ijk} (e_2)_j (e_3)_k \varepsilon^{mny} (e^2)_n (e^3)_y = 1$$

$$\varepsilon^{ijk} (e_2)_j (e_3)_k \varepsilon^{iny} (e^2)_n (e^3)_y = 1$$

$$(\delta_n^j \delta_y^k - \delta_y^j \delta_n^k) (e_2)_j (e_3)_k (e^2)_n (e^3)_y = 1$$

$$\delta_n^j \delta_y^k (e_2)_j (e_3)_k (e^2)_n (e^3)_y - \delta_y^j \delta_n^k (e_2)_j (e_3)_k (e^2)_n (e^3)_y = 1$$

$$(e_2 \cdot e^2)(e_3 \cdot e^3) - (e_2 \cdot e^3)(e_3 \cdot e^2) = 1 \Rightarrow 1 \cdot 1 - 0 = 1 \Rightarrow 1 - 0 = 1 \Rightarrow 1 = 1$$

c) se busca un q tal que

$$q \cdot e^i = 1$$

de modo que para cumplir esa igualdad se necesita q

$$e_i \cdot e^i = \delta_i^i = 1$$

de este modo se evidencia la unicidad del vector $q = e_i$ donde como vector sus componentes son $q = 1$

d) se tiene

$$w_1 = 4i + 2j + k$$

$$w_2 = 3i + 3j$$

$$w_3 = 2k$$

se supone la base dual $(a_1, b_1, c_1) \in V^*$ que cumple con $\langle e^i | e_j \rangle = \delta_j^i$

$$\langle e^1 | e_1 \rangle = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} \frac{1}{\sqrt{21}} (4, 2, 1) = \frac{1}{\sqrt{21}} (4a_1 + 2b_1 + c_1) = 1$$

$$\langle e^1 | e_2 \rangle = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} \frac{1}{3\sqrt{2}} (3, 3) = \frac{1}{3\sqrt{2}} (3a_1 + 3b_1 + 0) = 0$$

$$\langle e^1 | e_3 \rangle = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} \frac{1}{2} (0, 0, 2) = \frac{1}{2} (0 + 0 + 2c_1) = 0$$

$$\langle e^1 | = \begin{pmatrix} \sqrt{21}/2 \\ -\sqrt{21}/2 \\ 0 \end{pmatrix}$$

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$$\langle e^2 | = \begin{pmatrix} -\sqrt{2}/2 \\ \sqrt{2} \\ 0 \end{pmatrix}$$

$$\langle e^3 | e_1 \rangle = \begin{pmatrix} a_3 \\ b_3 \\ c_3 \end{pmatrix} \frac{1}{\sqrt{21}} (4, 2, 1) = \frac{1}{\sqrt{21}} (4a_3 + 2b_3 + c_3) = 0$$

$$\langle e^3 | e_2 \rangle = \begin{pmatrix} a_3 \\ b_3 \\ c_3 \end{pmatrix} \frac{1}{3\sqrt{2}} (3, 3, 0) = \frac{1}{3\sqrt{2}} (3a_3 + 3b_3 + 0) = 0$$

$$\langle e^3 | = \begin{pmatrix} -1/2 \\ 1/2 \\ 1 \end{pmatrix}$$

se supone la base dual $(a_i, b_i, c_i) \in V^*$ que cumple con $\langle e_i | e_j \rangle = \delta_{ij}$

$$\langle e^1 | e_1 \rangle = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} \frac{1}{\sqrt{11}} (4, 2, 1) = \frac{1}{\sqrt{11}} (4a_1 + 2b_1 + c_1) = 1$$

$$\langle e^1 | e_2 \rangle = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} \frac{1}{3\sqrt{2}} (3, 3) = \frac{1}{3\sqrt{2}} (3a_1 + 3b_1 + 0) = 0$$

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$$\langle e^1 | = \begin{pmatrix} \sqrt{11}/2 \\ -\sqrt{11}/2 \\ 0 \end{pmatrix}$$

$$\langle e^2 | e_1 \rangle = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix} \frac{1}{\sqrt{11}} (4, 2, 1) = \frac{1}{\sqrt{11}} (4a_2 + 2b_2 + c_2) = 0$$

$$\langle e^2 | e_2 \rangle = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix} \frac{1}{3\sqrt{2}} (3, 3) = \frac{1}{3\sqrt{2}} (3a_2 + 3b_2 + 0) = 1$$

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$$\langle e^3 | = \begin{pmatrix} -1/2 \\ 1/2 \\ 1 \end{pmatrix}$$

III) Componentes contravariantes

$$V^i = V \cdot e^i$$

$$1 = (1, 2, 3) \cdot (\sqrt{2}/2, -\sqrt{2}/2, 0) \\ = \sqrt{2}/2 - \sqrt{2} + 0 = -\sqrt{2}/2$$

$$2 = (1, 2, 3) \cdot (-\sqrt{2}/2, \sqrt{2}, 0) \\ = -\sqrt{2}/2 + 2\sqrt{2} + 0 = 3\sqrt{2}/2$$

$$3 = (1, 2, 3) \cdot (-1/2, 1/2, 1) \\ = -1/2 + 1 + 3 = 7/2$$

Componentes covariantes

$$V_i = V \cdot e_i$$

$$1 = (1, 2, 3) \cdot (4, 2, 1) \\ = 4 + 4 + 3 = 11$$

$$2 = (1, 2, 3) \cdot (3, 3, 0) \\ = 3 + 6 + 0 = 9$$

$$3 = (1, 2, 3) \cdot (0, 0, 2) \\ = 0 + 0 + 6 = 6$$

7) se sabe que

$$\text{Tr}(\sigma_M^\dagger \sigma_V) = 2\delta_{MV} \quad \text{para } M, V = 0, 1, 2, 3 \quad \text{por ser hermitianos}$$

• la base reciproca debe cumplir

$$\text{Tr}((\sigma^M)^\dagger \sigma_V) = \delta_V^M$$

• Pero se tiene $\text{Tr}(\sigma_M^\dagger \sigma_V) = 2\delta_{MV}$ entonces se escala en $1/2$:

$$\sigma^0 = \frac{1}{2}\sigma_0 \quad \sigma^1 = \frac{1}{2}\sigma_1 \quad \sigma^2 = \frac{1}{2}\sigma_2 \quad \sigma^3 = \frac{1}{2}\sigma_3$$

• Por lo tanto

$$\text{Tr}((\sigma^1)^\dagger \sigma_1) = \text{Tr}\left(\frac{1}{2}\sigma_1 \cdot \sigma_1\right) = \frac{1}{2} \cdot 2 = 1 \quad M=V$$

$$\text{Tr}((\sigma^1)^\dagger \sigma_2) = \frac{1}{2} \text{Tr}(\sigma_1 \sigma_2) = 0 \quad M \neq V$$